

Angles

Name: _____

Date: _____

Year 6 Maths — Measurement

ACARA v9.0: AC9M6M04

Learning Intention

I am learning to identify the relationships between angles on a straight line, angles at a point, and vertically opposite angles.

Success Criteria

- I can calculate unknown angles on a straight line.
- I can calculate unknown angles at a point.
- I can identify and use vertically opposite angles.

3 Angle Rules

Angles on a straight line add to 180° . Angles at a point add to 360° . Vertically opposite angles (formed by crossing lines) are always equal.

Worked Example

Two angles on a straight line: one is 110° .
The other = $180^\circ - 110^\circ = 70^\circ$

Part A — Warm Up

Angles on a straight line add up to: **90°**
 180° **360°**

True or False: Vertically opposite angles are always equal. _____

Angles at a point add up to _____ $^\circ$.

Part B — Core Skills

Two angles on a straight line: one is 65° .
Find the other angle. _____

3 angles meet at a point: 120° , 95° and an unknown angle. Find the unknown angle.

Two lines cross, forming vertically opposite angles. One angle is 48° . Find its vertically opposite angle. _____

Check: do 3 angles on a straight line of 90° , 50° and 40° add correctly to 180° ? Show your working.

Word Bank

angle • degree ($^\circ$) • straight line • point • vertically opposite • sum • unknown angle

Working Space — Draw a Straight Line with 2 Angles and Label Their Sizes

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Part C — Challenge: The Skate Ramp Design

A group of Year 6 students in Geelong is designing a skate ramp. The ramp's surface meets the flat ground at one angle, and the flat ground continues in a straight line on the other side of the ramp.

Q8. If the ramp's angle with the ground is 35° , calculate the angle on the other side of the ramp along the straight ground line.

Q9. Two support beams cross underneath the ramp, forming vertically opposite angles. If one angle is 62° , find the other 3 angles formed by the crossing beams.

Q10. A student claims the 4 angles formed by the crossing support beams will always add up to 360° , no matter what angle the beams cross at. Explain whether this claim is correct, using angle rules to justify your answer.

Answer Key — Part A

A1: 180°

A2: True — vertically opposite angles are always equal.

A3: 360.

Answer Key — Part B

B1: 115° ($180 - 65 = 115$).

B2: 145° ($360 - 120 - 95 = 145$).

B3: 48° (vertically opposite angles are equal).

B4: Yes — $90 + 50 + 40 = 180^\circ$, so these 3 angles correctly add up to 180° and could be angles on a straight line.

Answer Key — Part C

C8: 145° ($180 - 35 = 145$).

C9: The angle vertically opposite 62° is also 62° , and the other two angles (adjacent to 62°) are each 118° ($180 - 62 = 118$), since angles on a straight line add to 180° .

C10: Correct — no matter the crossing angle, the 4 angles formed always consist of 2 pairs of equal vertically opposite angles, and adjacent angles always add to 180° along each straight line, meaning all 4 angles together always total 360° (angles at a point).

I Can Checklist

☐ I can calculate angles on a straight line.

☐ I can identify vertically opposite angles.

☐ I can calculate angles at a point.